

Learning a Codebase

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1 Statistics

1.1 Glossary - Summary

- The founder developer starts in the first six months of a project.
- Late joiner developers begin after six months.
- Transient developers have ten (10) or fewer commits.
- Moderate developers have more than ten (10) commits but fewer than 50 commits or commits for less than 250 days.
- Sustained developers make 50 or more commits and commit for 250 days or more.

1.2 Repository Summary Table

Table 1: Summary of fifteen open-source repositories identified from GitHub that have at least 1000 pull requests and at least zero sustained late joiner developers. Sustained later joiner developers joining the project after six months and contributed at least 50 commits over a period of 250 days or more. Please note that five developers each worked on two repositories.

ID	Repo Name	Transient Founder	Moderate Founder	Sustained Founder	Transient Joiner	Moderate Joiner	Sustained Joiner	Commit	Start	End
3	activiti	6	7	6	212	20	5	8662	2010-Jun-18	On going
5	airbyte-platform	16	2	1	457	32	6	6784	2020-Aug-04	2023-Sep-27
7	ambari	0	3	0	129	32	15	4247	2011-Sep-22	On going
10	automq	4	0	0	301	10	3	1874	2011-Aug-10	2017-Apr-21
14	buck	18	5	6	531	85	21	14876	2013-Mar-21	2022-May-05
15	camel	1	4	1	1146	63	13	36528	2007-Mar-19	On going
18	checkstyle	0	0	1	362	42	9	17338	2001-Jun-22	On going
20	cxfs	6	6	3	201	17	3	7893	2008-Apr-29	On going
21	intellij-community	0	3	7	344	93	84	101247	2004-Nov-11	2018-Apr-18
26	guava	0	1	0	402	8	3	9685	2009-Sep-15	On going
28	jenkins	5	1	1	837	39	4	30252	2006-Nov-05	On going
33	openmrs-core	0	1	0	388	26	3	15757	2006-May-03	On going
36	presto	0	1	3	692	71	18	30605	2012-Aug-09	On going
37	quarkus	2	3	5	824	25	8	81612	2018-Jun-14	On going
39	selenium	5	3	0	712	17	8	23354	2004-Nov-03	On going
Total		63	40	34	7538	580	203	390714		

1.3 Welch t test - For all components touched for all commits

A Welch t -test was employed to determine if the mean contributions of Group A differ significantly from Group B. This test is appropriate for continuous data and is robust to unequal variances. Complementarily, the Mann-Whitney U (MWU) test was used to assess stochastic dominance by ranking data points. This non-parametric approach does not require a normal distribution and is highly resistant to outliers. Columns below:

- **Sample A** - F - Founder, LJ - Late Joiner, T - Transient, M - Moderate, S Sustained. In brackets number of developers.
- **Sample B** - F - Founder, LJ - Late Joiner, T - Transient, M - Moderate, S Sustained. In brackets number of developers.
- **Component** - Packages, classes or method touched.
- **Number of first commits** - A sample of the first x number of commits for developers.
- **Sample A μ** - The mean average of the first sample A.
- **Sample B μ** - The mean average of the second sample B.
- **Welch Statistic** - This is the magnitude of the change, greater than 2.0 generally indicates a significant change.
- **Welch P** - A significance test of the difference between sample A and sample B. A $p < 0.05$ indicates a significant difference between samples.
- **MWU Statistic (Number of Pairs)** - The Mann-Whitney U Statistic represents the number of pairwise comparisons where one group outranks the other, with the total possible pairs provided in brackets.
- **MWU P** - The Mann-Whitney p -value. A significance test of the difference between sample A and sample B. A $p < 0.05$ indicates a significant difference between samples.

Table 2: Statistical significance tests of different categories of founder versus different categories of late joiner touches of different granularities of component (package, class, method) after 1, 5, 10 and 20 commits for contributors.

Sample A	Sample B	Component	Number of First Commits	Sample A μ	Sample B μ	Welch Statistic	Welch P	MWU Statistic (Number of Pairs)	MWU P
T,S,M & F(137)	T,S,M & LJ(8303)	Packages	1	0.44	0.10	2.76	0.0067	642,742.0 (1,137,511)	0.0000
M & F(40)	M & LJ(580)	Packages	1	0.80	0.51	0.77	0.4477	12,483.0 (23,200)	0.2044
S & F(34)	S & LJ(203)	Packages	1	0.38	0.65	-1.06	0.2921	3,461.5 (6,902)	0.9673
M,S & F(74)	M,S & LJ(783)	Packages	5	3.36	3.85	-0.99	0.3232	29,616.5 (57,942)	0.7459
M & F(40)	M & LJ(580)	Packages	5	3.52	3.72	-0.27	0.7874	12,075.0 (23,200)	0.6567
S & F(34)	S & LJ(203)	Packages	5	3.18	4.22	-1.47	0.1459	3,369.0 (6,902)	0.8231
M,S & F(74)	M,S & LJ(783)	Packages	10	5.07	6.05	-1.33	0.1855	28,593.0 (57,942)	0.8498
M & F(40)	M & LJ(580)	Packages	10	5.15	6.09	-0.95	0.3462	11,610.5 (23,200)	0.9925
S & F(34)	S & LJ(203)	Packages	10	4.97	5.94	-0.83	0.4084	3,298.5 (6,902)	0.6772
S & F(34)	S & LJ(203)	Packages	20	7.18	9.62	-1.41	0.1628	3,181.5 (6,902)	0.4616
T,S,M & F(137)	T,S,M & LJ(8303)	Classes	1	4.01	0.64	2.92	0.0042	744,396.0 (1,137,511)	0.0000
M & F(40)	M & LJ(580)	Classes	1	7.47	2.37	1.43	0.1603	12,255.5 (23,200)	0.4989
S & F(34)	S & LJ(203)	Classes	1	2.85	4.81	-0.76	0.4494	3,783.0 (6,902)	0.2850
M,S & F(74)	M,S & LJ(783)	Classes	5	39.05	24.11	1.77	0.0799	32,451.0 (57,942)	0.0872
M & F(40)	M & LJ(580)	Classes	5	41.38	22.04	1.70	0.0969	13,640.0 (23,200)	0.0626
S & F(34)	S & LJ(203)	Classes	5	36.32	30.01	0.48	0.6350	3,651.0 (6,902)	0.5895
M,S & F(74)	M,S & LJ(783)	Classes	10	64.81	43.51	2.07	0.0419	32,846.5 (57,942)	0.0569
M & F(40)	M & LJ(580)	Classes	10	61.60	41.95	1.40	0.1679	13,313.0 (23,200)	0.1180
S & F(34)	S & LJ(203)	Classes	10	68.59	47.98	1.30	0.2003	3,879.0 (6,902)	0.2478
S & F(34)	S & LJ(203)	Classes	20	107.50	81.91	1.34	0.1863	3,846.5 (6,902)	0.2857
T,S,M & F(137)	T,S,M & LJ(8303)	Methods	1	12.52	2.06	2.48	0.0144	743,810.0 (1,137,511)	0.0000
M & F(40)	M & LJ(580)	Methods	1	24.27	7.22	1.32	0.1957	12,272.0 (23,200)	0.4881
S & F(34)	S & LJ(203)	Methods	1	10.00	23.40	-0.77	0.4417	3,767.0 (6,902)	0.3092
M,S & F(74)	M,S & LJ(783)	Methods	5	164.59	86.68	1.56	0.1218	32,324.5 (57,942)	0.0995
M & F(40)	M & LJ(580)	Methods	5	186.75	76.85	1.46	0.1516	13,748.5 (23,200)	0.0499
S & F(34)	S & LJ(203)	Methods	5	138.53	114.77	0.36	0.7211	3,550.5 (6,902)	0.7890
M,S & F(74)	M,S & LJ(783)	Methods	10	263.26	154.41	1.76	0.0826	33,245.0 (57,942)	0.0358
M & F(40)	M & LJ(580)	Methods	10	271.43	145.21	1.28	0.2063	13,417.5 (23,200)	0.0973
S & F(34)	S & LJ(203)	Methods	10	253.65	180.70	1.00	0.3240	3,896.5 (6,902)	0.2291
S & F(34)	S & LJ(203)	Methods	20	401.24	295.10	1.16	0.2533	3,693.5 (6,902)	0.5131

Table 3: Statistical significance tests of M,F & LJ versus S,F & LJ touches of different granularities of component (package, class, method) after 1, 5 and 10 commits for contributors.

Sample A	Sample B	Component	Number of First Commits	Sample A μ	Sample B μ	Welch Statistic	Welch P	MWU Statistic (Number of Pairs)	MWU P
M,F & LJ(620)	S,F & LJ(237)	Packages	1	0.53	0.61	-0.43	0.6686	72,287.0 (146,940)	0.5695
M,F & LJ(620)	S,F & LJ(237)	Packages	5	3.71	4.07	-0.80	0.4217	70,136.5 (146,940)	0.2931
M,F & LJ(620)	S,F & LJ(237)	Packages	10	6.03	5.80	0.40	0.6905	73,098.0 (146,940)	0.9068
M,F & LJ(620)	S,F & LJ(237)	Classes	1	2.70	4.53	-0.91	0.3614	78,310.0 (146,940)	0.0871
M,F & LJ(620)	S,F & LJ(237)	Classes	5	23.29	30.92	-1.69	0.0919	74,229.5 (146,940)	0.8148
M,F & LJ(620)	S,F & LJ(237)	Classes	10	43.22	50.94	-1.51	0.1316	74,941.0 (146,940)	0.6500
M,F & LJ(620)	S,F & LJ(237)	Methods	1	8.32	21.48	-0.92	0.3564	78,436.0 (146,940)	0.0791
M,F & LJ(620)	S,F & LJ(237)	Methods	5	83.94	118.18	-1.48	0.1389	73,502.5 (146,940)	0.9921
M,F & LJ(620)	S,F & LJ(237)	Methods	10	153.35	191.16	-1.49	0.1366	73,214.0 (146,940)	0.9372

Statistical Models

Welch's t-test To specifically compare the two primary roles (**Sample A** vs. **Sample B**) without assuming equal variances or sample sizes, Welch's t -test is employed. The t -statistic is defined as:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}}} \quad (1)$$

Where:

- $\bar{X}_1, \bar{X}_2$ are the sample means of Sample A and Sample B.
- s_1^2, s_2^2 are the sample variances.
- N_1, N_2 are the sample sizes.

The degrees of freedom (ν) for this test are calculated using the Welch-Satterthwaite equation:

$$\nu \approx \frac{\left(\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}\right)^2}{\frac{(s_1^2/N_1)^2}{N_1-1} + \frac{(s_2^2/N_2)^2}{N_2-1}} \quad (2)$$

1.4 Anova generate - For all components touched for all commits

An Anova table for all components touched on average.

The Two-Way ANOVA evaluates the influence of two independent categorical variables—*Role* (Founder vs. Late Joiner) and *Tenure* (Moderate vs. Sustained)—on the number of unique architectural components touched. The analysis excludes and includes Transient developers to isolate the behaviors of contributors with established project engagement.

Explanation of Table Columns:

- **Sum Sq (Sum of Squares):** Represents the total variation explained by that specific factor. A higher Sum Sq relative to the Residual indicates that the factor has a strong influence on the outcome.
- **df (Degrees of Freedom):** The number of independent values that can vary in the calculation. For categorical variables, this is typically $N - 1$ levels.
- **F (F-statistic):** The ratio of the variance explained by the factor to the variance unexplained (Residual). A larger F-value suggests the factor is more likely to be significant.
- **p-Value ($PR(> F)$):** The probability that the observed differences occurred by chance. Using a significance threshold of $\alpha = 0.05$, values above this indicate no statistically significant effect.
- **ETA Sq** The effect size, quantifies the proportion of the total variance in your dependent variable.

Explanation of Table Rows:

- **Intercept:** Represents the baseline value of the dependent variable when all categorical factors are at their reference levels.
- **Role(F/L):** Tests the *Main Role Effect* of being a Founder versus a Late Joiner. The p-Value above 0.05 indicates that the timing of project entry does not significantly change the breadth of early-stage component exploration.
- **Category(M/S):** Tests the *Main Category Effect* of the eventual contributor category (Transient vs. Moderate vs. Sustained).
- **Interaction:** Tests the *Interaction Effect*. It evaluates if the effect of being a Founder is different for the different categories.
- **Residual:** Represents the “noise” or variance within the groups that cannot be explained by Role or Tenure.

1.4.1 Anova generate - For all components touched for a all commits excluding tranient

Table 4: Anova results for all components, number of commits and 8440 developers excluding transient

Component	Number of First Commits	Factor	Sum Squared	df	F-Stat	P Value	ETA Sq
packages	1	Role (F/LJ)	3.14	1	3.31	0.0688	0.0004
packages	1	Category (T/M/S)	7.86	2	4.15	0.0158	0.0010
packages	1	Interaction	5.65	2	2.98	0.0509	0.0007
packages	1	Residual	7995.45	8434	-	-	-
packages	5	Role (F/LJ)	1.43	1	0.05	0.8250	0.0001
packages	5	Category (M/S)	2.23	1	0.08	0.7826	0.0001
packages	5	Interaction	11.68	1	0.40	0.5279	0.0005
packages	5	Residual	24984.13	853	-	-	-
packages	10	Role (F/LJ)	33.04	1	0.59	0.4443	0.0007
packages	10	Category (M/S)	0.59	1	0.01	0.9185	0.0000
packages	10	Interaction	0.01	1	0.00	0.9889	0.0000
packages	10	Residual	48127.58	853	-	-	-
classes	1	Role (F/LJ)	973.61	1	24.93	0.0000	0.0029
classes	1	Category (T/M/S)	683.32	2	8.75	0.0002	0.0020
classes	1	Interaction	815.33	2	10.44	0.0000	0.0024
classes	1	Residual	329384.23	8434	-	-	-
classes	5	Role (F/LJ)	13981.95	1	7.35	0.0069	0.0082
classes	5	Category (M/S)	468.97	1	0.25	0.6198	0.0003
classes	5	Interaction	2774.71	1	1.46	0.2276	0.0016
classes	5	Residual	1623543.63	853	-	-	-
classes	10	Role (F/LJ)	14448.45	1	5.02	0.0254	0.0055
classes	10	Category (M/S)	897.52	1	0.31	0.5769	0.0003
classes	10	Interaction	15.03	1	0.01	0.9424	0.0000
classes	10	Residual	2457107.31	853	-	-	-
methods	1	Role (F/LJ)	10883.40	1	6.87	0.0088	0.0008
methods	1	Category (T/M/S)	8092.96	2	2.56	0.0777	0.0006
methods	1	Interaction	15271.02	2	4.82	0.0081	0.0011
methods	1	Residual	13353418.84	8434	-	-	-
methods	5	Role (F/LJ)	451979.70	1	8.27	0.0041	0.0093
methods	5	Category (M/S)	42733.87	1	0.78	0.3769	0.0009
methods	5	Interaction	121524.53	1	2.22	0.1364	0.0025
methods	5	Residual	46646591.43	853	-	-	-
methods	10	Role (F/LJ)	596144.39	1	7.60	0.0060	0.0085
methods	10	Category (M/S)	5808.58	1	0.07	0.7856	0.0001
methods	10	Interaction	46476.28	1	0.59	0.4417	0.0007
methods	10	Residual	66927848.79	853	-	-	-

Statistical Model

The differences between developer categories are assessed using a Two-Way ANOVA with interaction. The value for any individual developer metric (Y_{ijk}) is modeled as:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk} \quad (3)$$

Where:

- μ is the grand mean of the touched components.
- α_i represents the effect of the **Role** (Founder vs. Late Joiner).
- β_j represents the effect of the **Tenure** (Transient, Moderate, Sustained).
- $(\alpha\beta)_{ij}$ is the interaction effect between Role and Tenure.
- ϵ_{ijk} is the residual error.

The F -statistic for each factor is calculated by:

$$F = \frac{SS_{factor}/df_{factor}}{SS_{residual}/df_{residual}} \quad (4)$$

1.5 Tukey hsd - For all components touched for all commits

An Tukey Honestly Significant Difference table for all components touched on average.

Post-hoc Analysis (Tukey HSD): Following the ANOVA, a Tukey Honestly Significant Difference (HSD) test was performed. While ANOVA identifies if at least one group differs, Tukey HSD performs pairwise comparisons to identify specific differences between developer roles and tenure categories while controlling for Type I error rates across multiple comparisons.

Table 5: Significant Tukey HSD results for classes and number of commits, for 8440 developers

Component	Commits	Group A	Group B	Mean Diff	Adj <i>P</i> Value
classes	1	Founder Moderate	Founder Sustained	-4.62	0.0190
classes	1	Founder Moderate	Founder Transient	-5.05	0.0009
classes	1	Founder Moderate	Late Joiner Moderate	-5.10	0.0000
classes	1	Founder Moderate	Late Joiner Transient	-7.08	0.0000
classes	1	Late Joiner Moderate	Late Joiner Sustained	2.43	0.0000
classes	1	Late Joiner Moderate	Late Joiner Transient	-1.98	0.0000
classes	1	Late Joiner Sustained	Late Joiner Transient	-4.42	0.0000
classes	5	Founder Moderate	Late Joiner Moderate	-19.33	0.0345
classes	10	Founder Sustained	Late Joiner Moderate	-26.64	0.0258
methods	1	Founder Moderate	Late Joiner Transient	-23.19	0.0032
methods	1	Founder Transient	Late Joiner Sustained	16.99	0.0364
methods	1	Late Joiner Moderate	Late Joiner Sustained	16.18	0.0000
methods	1	Late Joiner Moderate	Late Joiner Transient	-6.14	0.0046
methods	1	Late Joiner Sustained	Late Joiner Transient	-22.32	0.0000
methods	5	Founder Moderate	Late Joiner Moderate	-109.90	0.0215
methods	10	Founder Moderate	Late Joiner Moderate	-126.22	0.0303

Table 6: Not Significant Tukey HSD results for classes and number of commits, for 8440 developers

Component	Commits	Group A	Group B	Adj <i>P</i>	
				Mean Diff	Value
classes	1	Founder Moderate	Late Joiner Sustained	-2.67	0.1341
classes	1	Founder Sustained	Founder Transient	-0.42	0.9996
classes	1	Founder Sustained	Late Joiner Moderate	-0.48	0.9981
classes	1	Founder Sustained	Late Joiner Sustained	1.95	0.5397
classes	1	Founder Sustained	Late Joiner Transient	-2.46	0.1975
classes	1	Founder Transient	Late Joiner Moderate	-0.05	1.0000
classes	1	Founder Transient	Late Joiner Sustained	2.38	0.0880
classes	1	Founder Transient	Late Joiner Transient	-2.04	0.1031
classes	5	Founder Moderate	Founder Sustained	-5.05	0.9599
classes	5	Founder Moderate	Late Joiner Sustained	-11.37	0.4344
classes	5	Founder Sustained	Late Joiner Moderate	-14.28	0.2486
classes	5	Founder Sustained	Late Joiner Sustained	-6.31	0.8631
classes	5	Late Joiner Moderate	Late Joiner Sustained	7.96	0.1137
classes	10	Founder Moderate	Founder Sustained	6.99	0.9443
classes	10	Founder Moderate	Late Joiner Moderate	-19.65	0.1135
classes	10	Founder Moderate	Late Joiner Sustained	-13.62	0.4581
classes	10	Founder Sustained	Late Joiner Sustained	-20.61	0.1631
classes	10	Late Joiner Moderate	Late Joiner Sustained	6.03	0.5137
classes	20	Founder Sustained	Late Joiner Sustained	-25.59	0.1034
methods	1	Founder Moderate	Founder Sustained	-14.28	0.6396
methods	1	Founder Moderate	Founder Transient	-17.86	0.2283
methods	1	Founder Moderate	Late Joiner Moderate	-17.05	0.0921
methods	1	Founder Moderate	Late Joiner Sustained	-0.88	1.0000
methods	1	Founder Sustained	Founder Transient	-3.59	0.9983
methods	1	Founder Sustained	Late Joiner Moderate	-2.78	0.9988
methods	1	Founder Sustained	Late Joiner Sustained	13.40	0.4546
methods	1	Founder Sustained	Late Joiner Transient	-8.92	0.7829
methods	1	Founder Transient	Late Joiner Moderate	0.81	1.0000
methods	1	Founder Transient	Late Joiner Transient	-5.33	0.8974
methods	5	Founder Moderate	Founder Sustained	-48.22	0.8132
methods	5	Founder Moderate	Late Joiner Sustained	-71.98	0.2840
methods	5	Founder Sustained	Late Joiner Moderate	-61.68	0.4410
methods	5	Founder Sustained	Late Joiner Sustained	-23.76	0.9470
methods	5	Late Joiner Moderate	Late Joiner Sustained	37.92	0.1931
methods	10	Founder Moderate	Founder Sustained	-17.78	0.9930
methods	10	Founder Moderate	Late Joiner Sustained	-90.73	0.2408
methods	10	Founder Sustained	Late Joiner Moderate	-108.44	0.1257
methods	10	Founder Sustained	Late Joiner Sustained	-72.95	0.4963
methods	10	Late Joiner Moderate	Late Joiner Sustained	35.49	0.4058
methods	20	Founder Sustained	Late Joiner Sustained	-106.14	0.1554

Post-hoc Analysis: Tukey HSD

Following the Two-Way ANOVA, a Tukey Honestly Significant Difference (HSD) test is used to conduct pairwise comparisons. This test determines which specific group means differ significantly while maintaining a constant family-wise error rate.

The studentized range statistic (q) for comparing two group means is defined as:

$$q = \frac{\bar{Y}_{max} - \bar{Y}_{min}}{\sqrt{\frac{MS_{residual}}{n}}} \quad (5)$$

Where:

- $\bar{Y}_{max}$ and $\bar{Y}_{min}$ are the larger and smaller means of the two groups being compared.
- $MS_{residual}$ is the Mean Square Error derived from the ANOVA residual.
- n is the number of observations in each group (adjusted for unequal sample sizes via the harmonic mean).

A pair of groups is considered significantly different if the absolute difference between their means exceeds the Honestly Significant Difference (HSD):

$$HSD = q_{\alpha, k, df} \sqrt{\frac{MS_{residual}}{n}} \quad (6)$$

Where $q_{\alpha, k, df}$ is the critical value from the studentized range distribution based on the significance level ($\alpha = 0.05$), the number of groups (k), and the degrees of freedom (df) from the ANOVA residual.

1.6 Strategy logistic regression - For all components touched for all commits

To identify early indicators of long-term developer engagement, a Logistic Regression model was employed. The model treats the developer category (Sustained vs. Non-Sustained) as a binary dependent variable. The following metrics are reported:

- **Coefficient (β):** This value represents the change in the log-odds of a developer becoming 'Sustained' for every unit increase in the component count. A positive coefficient indicates that higher activity in that component increases the likelihood of long-term retention.
- **Odds Ratio (e^β):** Perhaps the most interpretable metric, the Odds Ratio quantifies the relative increase in the likelihood of being a Sustained developer. For example, an Odds Ratio of 1.05 implies a 5% increase in the odds of becoming sustained for each additional component touched.
- **Std. Err (Standard Error):** Measures the accuracy of the coefficient estimate. A lower standard error relative to the coefficient indicates a more precise estimate.
- **p-Value ($P > |z|$):** Indicates the statistical significance of the predictor. In this study, a p-Value < 0.05 suggests that the specific component count is a significant predictor of whether a developer will transition into a sustained role.
- **50% Threshold:** The threshold number of components to touch to have a 50
- **Moderate Met Threshold:** The moderate number of developers to have met the threshold, the percentate of developers in brackets.
- **Sustained Met Threshold:** The sustained number of developers to have met the threshold, the percentate of developers in brackets.

Interpretation of Intercept vs. Component Count:

- **Intercept:** Represents the baseline log-odds of the dependent variable (Sustained Status) when the independent variable (e.g., Package Count) is zero. In this study, the consistently negative intercepts indicate that the probability of a developer becoming 'Sustained' is statistically low without active engagement in the codebase.
- **Component Count (Slope):** Represents the effect size of architectural engagement. The positive coefficients and Odds Ratios greater than 1.0 demonstrate that every additional unique component touched by a developer significantly increases their likelihood of transitioning into a sustained contributor.

Table 7: Strategy Logistic Regression for all components against number of commits for 8440 developers excluding trainsient

Component	Number of		Moderate Sustained		Coefficient	Odds Ratio	Std. Err	p -Value	Moderate Sustained		
	First	Type	μ	μ					30% Threshold	Met Threshold	Met Threshold
packages	1	Intercept	0.40	3.99	-3.8260	0.0218	0.0752	0.0000	-	-	-
packages	1	Packages Count	0.40	3.99	0.2270	1.2548	0.0166	0.0000	13.1243	29 (0.4%)	17 (7.2%)
packages	5	Intercept	7.36	16.53	-1.3025	0.2719	0.0978	0.0000	-	-	-
packages	5	Packages Count	7.36	16.53	0.0324	1.0329	0.0055	0.0000	14.0420	116 (18.7%)	80 (33.8%)
packages	10	Intercept	9.68	23.99	-1.4121	0.2436	0.1019	0.0000	-	-	-
packages	10	Packages Count	9.68	23.99	0.0309	1.0313	0.0044	0.0000	18.3064	123 (19.8%)	90 (38.0%)
classes	1	Classes Count	2.82	36.49	0.0419	1.0428	0.0026	0.0000	74.0370	44 (0.5%)	28 (11.8%)
classes	5	Intercept	56.38	240.08	-1.8719	0.1538	0.1205	0.0000	-	-	-
classes	5	Classes Count	56.38	240.08	0.0092	1.0092	0.0009	0.0000	111.4174	86 (13.9%)	125 (52.7%)
classes	10	Intercept	79.55	407.66	-2.5977	0.0744	0.1560	0.0000	-	-	-
classes	10	Classes Count	79.55	407.66	0.0107	1.0108	0.0009	0.0000	162.8939	66 (10.6%)	160 (67.5%)
methods	1	Methods Count	9.44	133.10	0.0084	1.0084	0.0006	0.0000	355.1713	34 (0.4%)	16 (6.8%)
methods	5	Intercept	207.63	926.61	-1.6702	0.1882	0.1087	0.0000	-	-	-
methods	5	Methods Count	207.63	926.61	0.0019	1.0019	0.0002	0.0000	438.0841	67 (10.8%)	113 (47.7%)
methods	10	Intercept	289.62	1574.80	-2.2477	0.1056	0.1361	0.0000	-	-	-
methods	10	Methods Count	289.62	1574.80	0.0022	1.0022	0.0002	0.0000	624.0901	63 (10.2%)	150 (63.3%)

Predictive Identification Model

To determine if early-stage developer behavior can identify future sustained contributors, we employ a Logistic Regression model. The probability P of a developer becoming a **Sustained Contributor** is modeled using the log-odds (logit) function:

$$\ln\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n + \epsilon \quad (7)$$

Where:

- P is the probability that the developer’s tenure is “Sustained”.
- β_0 is the intercept.
- X_n represents early-stage metrics (e.g., methods or classes touched in the first 10, 20, or 50 commits).
- β_n are the coefficients representing the weight of each predictor.

The impact of each behavior is quantified using the **Odds Ratio (OR)**, calculated as:

$$OR = e^{\beta_n} \quad (8)$$

An $OR > 1$ indicates that an increase in the metric (e.g., touching more classes) increases the likelihood of long-term retention.

Model Evaluation The predictive power is evaluated using the Area Under the Receiver Operating Characteristic Curve (AUC-ROC). An AUC of 0.5 represents random chance, while an AUC closer to 1.0 indicates high predictive accuracy.